Equipment data sheet: K-202

Status: DRAFT for engineering review, not approved. Generated by PlantAgent from the workspace record; every number carries its source.

Field	Value	Unit	Source
Plant	Vacuum-salt plant with vapour recompression evaporator (US 5,366,514 flowsheet)		cases/salt_mvr/case.json
Tag	K-202		proposal
Equipment class	compressor		model unit type
Status	proposed new equipment (answer 4: Add a small parallel vapour compressor K-202), rating to confirm		answer 4
Model unit	K-202		same tag in the model
Throughput (basis solve)	120	% of today	answer 4 proposal 'Add a small parallel vapour compressor K-202': calculation record (dwsim, 120 %, model vrc_cal_v2_desup_hx)

Process conditions (simulator)

Field	Value	Unit	Source
discharge_C	191.5	°C	basis solve (120 %)
duty_kW	2021	kW	basis solve (120 %)
kW	2021	kW	basis solve (120 %)
inlet S3001_K202 T_C	115.2	°C	basis solve (120 %)
inlet S3001_K202 P_bar	1.7	bar	basis solve (120 %)
inlet S3001_K202 kg_h	48750	kg/h	basis solve (120 %)
inlet S3001_K202 vapour_frac	1		basis solve (120 %)
outlet S3002_K202 T_C	191.5	°C	basis solve (120 %)
outlet S3002_K202 P_bar	3.08	bar	basis solve (120 %)
outlet S3002_K202 kg_h	48750	kg/h	basis solve (120 %)
outlet S3002_K202 vapour_frac	1		basis solve (120 %)

Design data

Field	Value	Unit	Source
P_out_bar	3.08	bar	model spec units.K-202.P_out_bar (proposal add_unit)
eff_pct	75	%	model spec units.K-202.eff_pct (proposal add_unit)

Ratings (limits the checks use)

Rating	Value	Unit	Source
max_kW	2400	kW	vendor quote to confirm / standard size 55 t/h suction, 2,400 kW shaft at pressure ratio 1.8 (3.08 bar discharge) - proposed rating only, not an existing rating
max_kg_h	55000	kg/h	vendor quote to confirm / standard size 55 t/h suction, 2,400 kW shaft at pressure ratio 1.8 (3.08 bar discharge) - proposed rating only, not an existing rating

Limit checks at this condition

Check	Value	Limit	Utilisation	Source
suction flow vs capacity	48750 kg/h	55000 kg/h	88.6 %	basis solve (120 %)

shaft power vs motor	2021 kW	2400 kW	84.2 %	basis solve (120 %)

To confirm

• [ ] Vendor curve and guaranteed performance

• [ ] Mechanical design and materials of construction

• [ ] Relief, isolation and control design (not on the as-built P&ID)

• [ ] Rating: vendor quote to confirm / standard size 55 t/h suction, 2,400 kW shaft at pressure ratio 1.8 (3.08 bar discharge) - proposed rating only, not an existing rating